SINGAPORE DIMENSIONS • GRADE 3A • CHAPTER 3

Multiplication & Division: Arrays & Groups

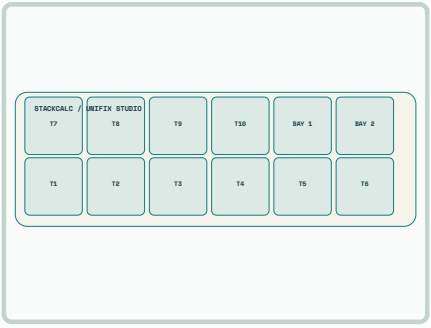
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.OA.A.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Arrange Counting & Array tiles into 4 rows of 6 columns on the array stage.

Sketch Model on Paper:
Sketch a rectangular area array labeled 4 and 6.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	7	X: 7	Key 1st operand
2.	ENTER	Y: 7 | X: 7	Stack lift copies X
3.	4	Y: 7 | X: 4	Key 2nd operand
4.	×	X: 28	Drop stack & compute

EXPECTED LCD X: 28 desks

1 6 rows of 8 chairs. Total chairs?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 9 bags with 5 apples each. Total apples?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 36 stickers shared equally by 4 students.

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 42 markers divided into boxes of 6.

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 3

Multiplication & Division: Arrays & Groups • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01)

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/counting_array_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Case & 4-Color Array Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**6 rows of 8 chairs. Total chairs?** Standard: CCSS.MATH.3.OA.A.3	48	[6] [ENTER] [8] [×]
2	**9 bags with 5 apples each. Total apples?** Standard: CCSS.MATH.3.OA.A.3	45	[9] [ENTER] [5] [×]
3	**36 stickers shared equally by 4 students.** Standard: CCSS.MATH.3.OA.A.3	9 stickers each	[3][6] [ENTER] [4] [÷]
4	**42 markers divided into boxes of 6.** Standard: CCSS.MATH.3.OA.A.3	7 boxes	[4][2] [ENTER] [6] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Division is the inverse: total in Y, group size in X, [÷] yields row count.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.